Final Examination
ECO 101: Principles of Microeconomics
Department of Economics
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Monsoon 2018.

This is a two hour long closed book, closed notes in class examination. Answer all the questions, and be clear and precise in your answers. Any act of *academic dishonesty* will be taken very seriously. *All the best.*

Question 1 (50 points)

State whether each of the following statements is True or False, and why so. Explain your findings precisely, with clear reasoning, and graphical and or mathematical illustration, wherever necessary.

(a.) A firm operates under the production function: $q = \min\{2L, 3K\}$, where L and K are the labour and capital employed in the production. The wage and rental rate of labour and capital is given by 1 and 5 respectively. The cost function of the firm (production cost as a function of output) is linear in output. True/False

(b.) A monopoly, who faces demand curve $P(Q) = 100 - Q$ charges a price 50 and therefore produces only 50 units of output. At the monopoly equilibrium, the price elasticity of demand turns out to be 0.93. True/False

(c.) A Giffen good is an inferior good, but an inferior good is not necessarily a Giffen good. True/False. [Hint: Just draw a graph to explain.]

(d) A firm in a perfectly competitive market have the cost curve that is given by: $C(q) = 200 + aq - bq^2 + q^3$. What is the price at which the firm no longer continues to produce and thus shuts down?

(e.) A stackelberg leader will necessarily earn a higher profit than a Cournot firm. True/False

Question 2 (50 points)

Consider a hypothetical town, Gotham City, with the following demand for drinking water. In Gotham city, only two residents, Bruce Wayne and Bane owns wells that produce water safe to drink. The cost of pumping water from wells is zero.
Initially, only Bruce was pumping water from his well in order to sell in Gotham city. Bane was using his well to use the water himself.

(a) When Bruce is the only seller of water, how much water will he pump and bring to the market? What price will he charge for the water per gallon? how much profit Bruce earns from selling water in Gotham city. (10 points)

(b) To have a benchmark, find out how many gallons of water will be sold in the market if the market was perfectly competitive. Find out the competitive price and profit each seller earns in the market. (5 points)

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TABLE 1. Market for Water in Gotham City

Quantity (Q)	Price P	Total Revenue $P * Q$
0	120	0
10	110	1100
20	100	2000
30	90	2700
40	80	3200
50	70	3500
60	60	3600
70	50	3500
80	40	3200
90	30	2700
100	20	2000
110	10	1100
120	0	0

Knowing that Bruce is making a handsome amount of money selling water, Bane decided to pump water from his well and sell in the market.

(c) Formulate the situation as a normal form game, carefully indicating the strategy space and pay-off structure. (10 points)

Since they know each other, Bane decided to have a chat before going to the market to sell water and compete with Bruce. When Bane met Bruce, Bruce came up with an idea: "why don't we together serve the market as a monopoly, equally divide what I was selling in the market when I was a monopoly and charge the monopoly price"said Bruce, "that way, each of us will earn half of monopoly profit and therefore will not have to compete with each other". Bane agreed, and went home.

(d) Does Bane have an incentive to stick to the verbal commitment he made to Bruce. Draw the relevant pay-off matrix, and explain whether there exists a unilateral incentive to deviate for Bane. (10 points)

(e) Assuming that Bane will abide by his commitment, what is the best response strategy to play for Bruce? (5 points)

(f) As a result of such a strategic interaction, which strategy profile emerges as a Nash equilibrium? How much each of them earn as profit in the Nash equilibrium? Explain the outcome of the game and precisely point out the forces that drove both the parties to that outcome. (10 points) ?

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